

When Should an LLM Search?

A Calibration-Centric Evaluation of Search Behavior and a Hidden Generalization Gap Under Natural Paraphrases

Anonymous Authors

Paper under double-blind review at ICLR 2026

Abstract

Modern LLMs are increasingly deployed as info-seeking assistants equipped with web-search tools — the canonical setting in which a user asks the model to find, verify, or assemble facts and the model must decide, on the fly, whether to lean on its parameters or to reach out to the live web. As frontier systems improve in parallel along two axes — richer parametric knowledge and stronger agentic tool use — the desired interplay between recalling and retrieving on a given info-seeking query becomes genuinely unclear: every emitted search trades latency, dollars, and the risk of context overriding a correct parametric belief against the chance to confirm or patch one. Yet we still lack a principled methodology for asking the most basic question about this behavior: *given that a model has the option to search, does it search when it should, and skip when it shouldn't?* Existing evaluations measure end-task accuracy or whether-or-not to use a tool at the question level; they do not localize search decisions to specific reasoning steps, nor do they reference the model’s own internal knowledge as the natural decision boundary. We close this gap with an evaluation protocol on 600 staleness-filtered MuSiQue (Trivedi et al., 2022) multi-hop questions. We first estimate per-hop *parametric uncertainty* via semantic entropy (Farquhar et al., 2024) over 5 closed-book samples, then run the same model on the full question with web search enabled, and use an LLM judge to attribute each emitted search query back to a specific hop. The cross-product of “certain vs. uncertain” and “searched vs. not searched” — refined with a sequential-redundancy flag and a cross-hop-coverage flag — partitions every per-hop decision into six interpretable behavioral categories: three correct decisions (*Necessary search*, *Correct skip*, *Cross-hop covered*) and three search errors (*Truly missed*, *Sequentially redundant*, *Parametrically redundant*). Applied to Gemini 3.1 Pro, Qwen 3.5 (122B) and Nemotron Nano (30B), the extended profile exposes three sharply different failure styles invisible to aggregate metrics. We then show the central calibration finding: the AUROC of entropy as a predictor of the search decision sits in the 0.56–0.62 range across all three models — search is barely aligned with the model’s own uncertainty. Finally, we introduce a paraphrase probe: every question is rewritten into a natural, user-styled question that preserves the underlying information need but removes the benchmark’s explicit hop chain. All three models search substantially less under the natural rewrite, and the searches they retain are re-allocated *away from* the uncertain hops the same model had identified on the benchmark. The dominant new category is *Truly missed* — a hop the model is internally uncertain about and never queries the web for — exposing a hidden generalization gap that existing benchmarks do not measure.

1 Introduction

Info-seeking queries — requests where a user asks the model to find, verify, or assemble facts — are the canonical setting in which modern LLMs are paired with web-search tools, and retrieval-augmented LLMs are now a default deployment pattern for them (Huang and Huang, 2024). Two simultaneous trends make the right interplay between parametric recall and live retrieval on these queries increasingly opaque. First, frontier models keep gaining parametric knowledge, so for any given fact the chance that the answer is already inside the model is non-negligible and growing. Second, the same models have grown markedly more agentially

capable, so emitting additional search calls is essentially free at the policy level — but not free in latency, dollars, or the risk that retrieved context overrides a correct parametric belief (Wu et al., 2024a; Tao et al., 2024). There is no consensus, theoretical or empirical, on what the *right* division of labor between parametric memory and live search should look like in this regime; in particular, neither “always search” nor “search only when the closed-book answer is wrong” is operationally well-defined when the model is a black box with no exposed knowledge boundary.

The standard evaluation question is whether retrieval improves end-task scores; a far less studied question is whether the model uses retrieval *appropriately* on these info-seeking queries — i.e. whether it triggers searches on facts it does not know and skips them on facts it does. Tool-use benchmarks such as WTU-Eval (Ning et al., 2024) frame the problem at the granularity of the entire question (use the tool or not), and the broader RAG / tug-of-war literature (Cheng et al., 2024a; Wu et al., 2024a; Tao et al., 2024; Kim et al., 2025; Farahani and Johansson, 2024) focuses on how models *integrate* external evidence once retrieved. Per-step retrieval gating has been studied in pipeline-RAG settings (Jiang et al., 2023; Asai et al., 2024; Yao et al., 2025; Guan et al., 2025; Wu et al., 2025a), and recent work shows that even these binary-gate decisions are brittle to surface-form variation (Jang et al., 2026). Our work extends this analysis to a qualitatively different setting: a multi-hop *agentic* info-seeking system with live web-search tool access, where the agent must decide *what* to query, *when*, and whether to continue, uncertainty propagates across reasoning hops, and the evaluation is grounded in per-hop parametric uncertainty rather than a corpus-retrieval gate or outcome correctness.

This is the question we address. Crucially, we deliberately do *not* ground our analysis in end-task correctness. The premise of the paper is that an external retrieval decision should be aligned with the model’s own internal epistemic state — regardless of whether the model would, downstream, happen to produce the right final answer. A model that searches questions it knows perfectly and skips questions it doesn’t is *mis-calibrated* as a search agent, independently of whether retrieval rescues the final answer.

Contributions.

1. An **evaluation protocol** that operationalises the parametric↔search interplay on agentic info-seeking queries: per-hop semantic entropy (the model’s own internal knowledge boundary, from closed-book sampling) paired with per-hop search attribution (the model’s revealed retrieval decision, from an LLM judge) on multi-hop questions (§3).
2. A **six-category extended search-calibration profile** that makes the interplay legible. The protocol partitions every per-hop decision into three correct cells and three distinct error cells; the correct cells distinguish *Cross-hop covered* (uncertain, not directly searched but answered as a by-product of another hop’s search) from *Truly missed* (uncertain, never searched and never covered) on the error side, and the error cells separate *Sequentially redundant* (re-searching already-known content within a trace) from *Parametrically redundant* (searching content the model already knows parametrically) — the two distinct ways a model can mis-allocate the search budget relative to its own parametric knowledge (§4).
3. A **cross-model calibration analysis** on three production-style LLMs showing (i) the six-category profile yields three sharply different parametric↔search archetypes — a wasted-search-heavy searcher, a coverage-failing under-searcher, and an intermediate profile — and (ii) the AUROC of entropy → search sits near chance for all three: search behaviour on these info-seeking queries is barely driven by the model’s own knowledge boundary, the headline diagnostic that the interplay is mis-calibrated (§4,5).
4. A **natural-paraphrase generalization probe** showing that a cosmetic rephrasing that preserves the underlying information need causes all three models to search less — and, more strikingly, to redirect their remaining searches *away from* their own uncertain hops, with *Truly missed* as the dominant new error cell. Surface framing of an info-seeking query, not internal uncertainty, ends up governing the parametric↔search policy (§6).

We argue that the paraphrase result is the most important finding of the paper. It is consistent with the literature on compositional brittleness (Li et al., 2024; Yang et al., 2024; Wu et al., 2024b) and with recent findings that framing shifts retrieval-gate decisions in pipeline-RAG systems (Jang et al., 2026), but extends both to an *agentic info-seeking* setting: even when the underlying parametric knowledge has not changed, a

cosmetic paraphrase causes the model to re-balance the parametric $\leftrightarrow$ search interplay in the wrong direction, with the mis-allocation traceable to specific reasoning hops.

2 Related Work

Tool-use and search-decision evaluation. WTU-Eval (Ning et al., 2024) measures tool-use at the *question level*. Step-level retrieval gating has been studied extensively in *pipeline-RAG* settings: FLARE (Jiang et al., 2023) triggers retrieval on low-probability tokens; Self-RAG (Asai et al., 2024) learns inline reflection tokens; DeepRAG (Guan et al., 2025) formalises per-subquery retrieval as an MDP; HiPRAG (Wu et al., 2025a) assigns hierarchical process rewards per reasoning step; ReaLM-Retrieve (Guo et al., 2026) introduces a per-step Reasoning-Step Uncertainty Signal tested on MuSiQue. SMART-ER (Qian et al., 2025) and Wu et al. (2025b) define over-search and under-search rates as explicit calibration metrics. Toolformer (Schick et al., 2023) and RARE (Wang et al., 2025b) train search insertion during generation but evaluate downstream task accuracy rather than per-decision calibration. All of this work operates over fixed document-retrieval pipelines; the search decision is a binary switch, not a free-form query to a live web-search tool inside an iterative reasoning loop. Our work evaluates this qualitatively different, harder setting — agentic info-seeking with a live web tool — and grounds calibration in per-hop parametric uncertainty rather than outcome correctness.

Parametric vs. contextual knowledge — the interplay that motivates this paper. A growing line studies how LLMs balance internal priors against retrieved context (Cheng et al., 2024a; Wu et al., 2024a; Farahani and Johansson, 2024; Kim et al., 2025; Wadhwa et al., 2026; Tao et al., 2024; Zhang et al., 2024). These works probe *conflict resolution* — what happens when internal and external knowledge disagree — rather than the upstream question of *when to invoke retrieval at all* on an info-seeking query. More recent work has moved toward per-token conflict arbitration: AdaCAD (Wang et al., 2025a) measures per-step Jensen-Shannon divergence between parametric and contextual distributions; CoCoA (Khandelwal et al., 2025) uses a composite per-step signal (Rényi divergence, entropy gap, contextual peakedness) to gate which logits dominate; Micro-Act (Huo et al., 2025) forces an explicit intermediate self-verification step before committing to a retrieved fact. We treat all of these as direct motivation: collectively they show the parametric $\leftrightarrow$ contextual interplay is fragile even *after* retrieval has been triggered — context can override correct parametric knowledge at the token level. The natural consequence is that on info-seeking queries the *upstream* decision — whether to retrieve at all, given that the model may already have the answer parametrically — is itself a load-bearing calibration problem, not a free action. As frontier models continue to gain parametric knowledge in parallel with stronger agentic tool use, the desired shape of this upstream policy becomes increasingly unclear, and yet has remained largely unmeasured at hop granularity in the agentic setting.

Uncertainty quantification. We use semantic entropy (Farquhar et al., 2024; Kunitomo-Jacquin et al., 2026) as our operational definition of “knows / does not know.” Semantic entropy correlates well with hallucination (Farquhar et al., 2024) and is cheaper than fine-grained atomic factuality (Min et al., 2023); Semantic Entropy Probes (Kossen et al., 2024) further reduce cost to a single forward pass via hidden-state linear probes. SEARAG (SEARAG, 2025) applies per-hop semantic entropy gating inside a pipeline-RAG multi-hop setting — the closest prior work in spirit to our per-hop uncertainty measurement; we extend this to the agentic setting and use entropy as a *post-hoc diagnostic* rather than a training-time gate.

Multi-hop QA and compositional reasoning. MuSiQue (Trivedi et al., 2022) supplies the hop-decomposed questions we need; MINTQA (He et al., 2024) probes the boundary between parametric memorization and genuine retrieval-dependent reasoning with gold sub-question annotations. The compositional-brittleness literature (Li et al., 2024; Yang et al., 2024; Wu et al., 2024b) documents how multi-hop performance collapses with hop-count and under surface-form variation: CREME (Li et al., 2024) shows that a repaired model must survive semantically equivalent paraphrases of the same two-hop chain. We add: search *behavior* in agentic systems is brittle to surface form independently of reasoning accuracy. For trace-level evaluation methodology we follow Lee and Hockenmaier (2025); the validity of LLM-judge per-step attribution is

supported by Han et al. (2025), who find that models $\geq 70\text{B}$ reach or exceed human-to-human inter-rater agreement on step-level annotation tasks.

Framing and paraphrase sensitivity. Framing sensitivity in retrieval-pipeline systems has recently been documented: Jang et al. (2026) measure retrieval-decision flip rates of up to 55% under meaning-preserving paraphrases in adaptive RAG; Cheng et al. (2026) formalise the “knowing-doing gap” — a mismatch between a model’s internal recognition that a tool is needed and its actual generation of tool-call tokens, amplified under conversational framing; Rabinovich et al. (2025) document output-mode collapse (structured tool-call tokens dissolving into prose) under prompt rephrasing in function-calling settings. Our work extends this line to a multi-hop *agentic* context: by coupling surface-form variation with per-hop uncertainty tracking we identify *which specific reasoning hops* are abandoned under natural framing — a diagnostic that pipeline-level analyses cannot provide.

Action calibration. Standard calibration asks whether expressed confidence correlates with answer correctness (Farquhar et al., 2024). A complementary notion — *action calibration* — asks whether the decision to invoke a tool is aligned with the model’s actual epistemic state. Wang et al. (2025c) formalise this as the Decision-Knowledge Alignment Principle: tool-use decisions are miscalibrated whenever the decision boundary diverges from the internal knowledge boundary. TARG (Wang et al., 2026) and SUGAR (Zubkova et al., 2025) calibrate retrieval gates using logit margins and semantic entropy respectively; SeaKR (Yao et al., 2025) uses hidden-state Gram determinants. Our protocol is a *diagnostic* complement to these training-time and inference-time gating methods: rather than imposing a threshold, we measure post-hoc how well search decisions of three production models align with their parametric uncertainty at hop granularity.

3 Protocol

We evaluate a model M on a dataset of multi-hop questions $Q = \{q_i\}$, where each q_i decomposes into an ordered chain of sub-questions (hops) $h_{i,1}, \dots, h_{i,k_i}$ with gold answers $a_{i,j}$. The protocol has four stages.

Stage 0 — Staleness filtering. MuSiQue contains time-sensitive questions whose gold answers may have changed since the dataset’s release (e.g. office-holders, populations). Such questions confound a search-behavior study because the correct action — search to refresh — depends on external state, not on internal knowledge. We classify each question with a `gpt-oss:20b` LLM judge that flags whether the gold answer is plausibly time-dependent. The classification is a straightforward categorical judgment over two exhaustive and mutually exclusive classes (stale: current office-holders, statistics, records, ongoing relationships; not stale: historical events, biographical facts, geographic constants — see Appendix E for the full prompt). LLM judges handle such enumerated categorical tasks reliably (Lee and Hockenmaier, 2025; Han et al., 2025). From an initial pool we retain **600 non-stale questions** as our evaluation set.

Stage 1 — Per-hop parametric uncertainty. For each example q_i and each hop $h_{i,j}$, we sample the model $N = 5$ times *without search access*, recording the free-text answer. A separate clustering agent (`gpt-oss:120b`) groups the 5 answers into semantic equivalence classes following Farquhar et al. (2024), who establish that LLM-based semantic clustering reliably separates distinct meaning-equivalence classes and is preferable to surface-form string matching for free-text QA answers. We compute *semantic entropy* in bits over the cluster distribution.

We deliberately use sampling-based semantic entropy rather than logit-based uncertainty for two reasons. First, *API constraint*: Gemini 3.1 Pro does not expose per-token log-probabilities through its public API, making logit-based measures unavailable for one of our three models and thus incomparable across the evaluation. Second, *calibration*: RLHF post-training systematically sharpens token probability distributions, causing logit entropy to underestimate model uncertainty even when the model is genuinely ignorant — an effect documented for adaptive-retrieval gates (Jiang et al., 2023) and more broadly in the calibration literature (Farquhar et al., 2024). Sampling-based semantic entropy, which measures dispersion over the output-meaning distribution rather than over the next-token distribution, is robust to this sharpening. Because $N = 5$ admits

only seven partition signatures, semantic entropy takes one of seven discrete values:

$$H \in \{0, 0.722, 0.971, 1.371, 1.522, 1.922, 2.322\} \text{ bits},$$

corresponding to partitions $[5], [4,1], [3,2], [3,1,1], [2,2,1], [2,1,1,1], [1,1,1,1,1]$. We say a hop is **certain** if its semantic entropy is 0 and the closed-book sampling cluster is the gold-equivalent one. We say it is **uncertain** otherwise. The conjunction is important: entropy-only certainty would admit the failure mode of confident-wrong, where the model returns the same incorrect answer 5 times.

Stage 2 — Full-question, search-augmented run. We prompt the model with the *aggregate* multi-hop question and a web-search tool (Brave Search), allowing it to interleave search calls and reasoning freely. We record the full trace (every assistant turn, tool call, tool response) and the final answer. We choose Brave Search for *provider independence*: Brave operates its own independent web index with no product or training-data relationship with any of the three evaluated model vendors (Google, Alibaba, NVIDIA), eliminating the possibility that a model gains an advantage from familiarity with the search backend’s index structure or result format.

Stage 3 — Query-to-hop attribution. The aggregate trace contains a sequence of search queries $\{Q_1, \dots, Q_T\}$. We use `gpt-oss:120b` as an LLM judge to attribute each query to one of the k_i hops, using the hop questions, the query string, and the preceding reasoning as context (full prompt in Appendix E). This approach is supported by the methodology of Lee and Hockenmaier (2025) for step-level trace evaluation; Han et al. (2025) further show that models $\geq 70\text{B}$ reach or exceed human-to-human inter-rater agreement on step-level annotation tasks, validating `gpt-oss:120b` as a reliable annotator for this classification. The judge returns a confidence label and may flag a query as “aggregate / general” if it does not cleanly serve one hop; we additionally verify robustness by ablating on high-confidence-only attributions, finding qualitatively identical patterns. After attribution, each hop carries a binary `searchedi,j` $\in \{0, 1\}$ and a count `queries_assignedi,j`. Two trace-level annotations enrich the binary search flag:

- **seq_redundant** — the search for this hop is sequentially redundant: an earlier query in the same trace already returned the same content.
- **missed_search_answer_found_cross_hop** — although no query was directly attributed to this hop, a query attributed to a different hop returned the gold-equivalent content for this hop (the hop was *covered* as a by-product). Concretely, for each hop with `searchedi,j` = 0, we use `gpt-oss:120b` to check whether the gold answer for hop j appears in the search results of any query attributed to a different hop $j' \neq j$ in the same trace; the flag is set when the judge confirms a gold-equivalent match.

Models. We evaluate **Gemini 3.1 Pro** (Google, proprietary, frontier-scale), **Qwen 3.5 122B** (Alibaba, open-weight, large), and **Nemotron Nano 30B** (NVIDIA, open-weight, mid-size). The selection is guided by four criteria. (i) *Vendor independence*: the three models come from entirely separate organisations with independent pre-training corpora, post-training recipes, and tool-use training data; behavioral patterns shared across all three are therefore attributable to the phenomenon rather than to a common training artifact. (ii) *Scale diversity*: the models span roughly one order of magnitude in parameter count ($\sim 30\text{B}$, 122B , frontier-scale), letting us check whether the miscalibration pattern is scale-dependent. (iii) *Access modality*: one proprietary API-only model and two open-weight models served locally, spanning the proprietary/open divide relevant for reproducibility. (iv) *Native tool-use support*: all three have been explicitly post-trained for structured tool-call generation, a prerequisite for the protocol — models that cannot issue search queries cannot be evaluated on search decisions. Non-Gemini models are served via Ollama; we do not fine-tune any model, making this a zero-shot evaluation of pretrained behavior.

4 Extended Search-Calibration Profile per Model

Cross-tabulating the per-hop annotations from Stages 1–3 partitions every per-hop decision into one of six interpretable behavioral categories (Table 1). Three are correct decisions; three are search errors with distinct semantics.

Table 1: The six extended search-calibration categories. The first three are correct decisions; the remaining three are distinct error modes. “Searched” refers to whether the LLM-judge attributed at least one search query to the hop. **seq_red** = sequentially redundant within the trace; **x-hop** = the gold-equivalent content was returned by a query attributed to a different hop.

Category	Certain?	Searched?	seq_red	x-hop	Reading
Necessary search	no	yes	—	—	Correct: search was warranted
Correct skip	yes	no	—	—	Correct: model knew, did not waste a call
Cross-hop covered	no	no	—	yes	Correct: hop recovered via another hop’s query
Truly missed	no	no	—	no	Error: gap left open
Seq. redundant	yes	yes	yes	—	Error: re-querying already-known content
Par. redundant	yes	yes	no	—	Error: searching a hop the model knew

Figure 1 compresses the entire main result of the paper into a single chart: the four-cell (E / PR / M / CP) collapse of the six-category profile, shown per model and per dataset side by side. Reading left-to-right, the first three bars are the three models on benchmark MuSiQue; the next three are the same models on the natural paraphrase. The visual jumps that the rest of the paper unpacks are already visible here: the red (Missed) segment grows substantially in every natural-rewrite bar, the green (Effective) segment shrinks correspondingly, and the orange (Parametrically Redundant) segment contracts as overall search volume falls — without a compensating gain in coverage.

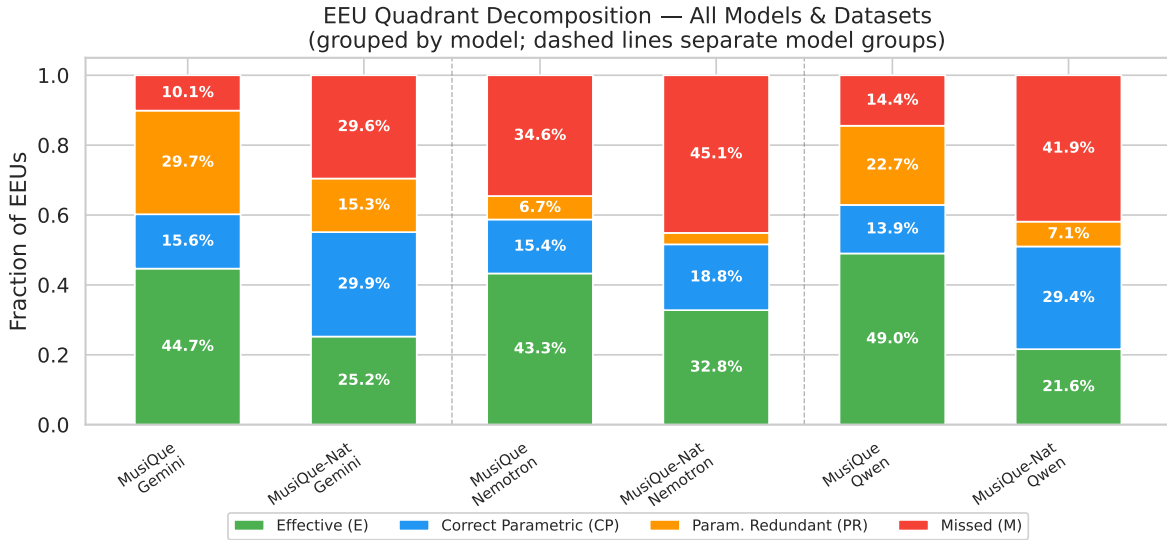


Figure 1: **Headline figure.** Four-cell taxonomy collapse of the six-category profile (E = Effective / Necessary search, CP = Correct Parametric / Correct skip, PR = Parametrically Redundant, M = Missed) per model and per dataset. Each bar sums to 100% of that (model, dataset)’s 1800 hops. Across all three models, the natural-paraphrase bars show a sharp expansion of the red (Missed) segment and a contraction of the green (Effective) segment — the central result of the paper.

Table 2 gives the full six-category breakdown per model on the benchmark. The extended categories distinguish two flavours of redundancy (sequential vs. parametric) and isolate the Cross-hop covered category, which would otherwise be conflated with Missed even though the hop’s information need is in fact satisfied.

Figure 2 shows the full extended-calibration plot for one representative model (Gemini 3.1 Pro), with the benchmark and natural-paraphrase bars overlaid per category. The remaining two per-model plots are in Appendix A; they tell the same story with different baseline distributions, as Table 4 in §6 quantifies.

Three model archetypes. The category mix reveals qualitatively different calibration failures:

Table 2: Extended search-calibration profile per model on MuSiQue (% of 1800 hops). Left of the rule: correct decisions. Right: distinct error modes.

Model	Necessary search	Correct Correct skip	Cross-hop covered	Truly missed	Errors Seq. redundant	Par. redundant
Gemini 3.1 Pro	44.7	15.6	5.1	5.1	9.2	20.5
Nemotron Nano 30B	43.3	15.4	10.2	24.3	1.3	5.4
Qwen 3.5 122B	49.0	13.9	7.7	6.7	2.8	19.8

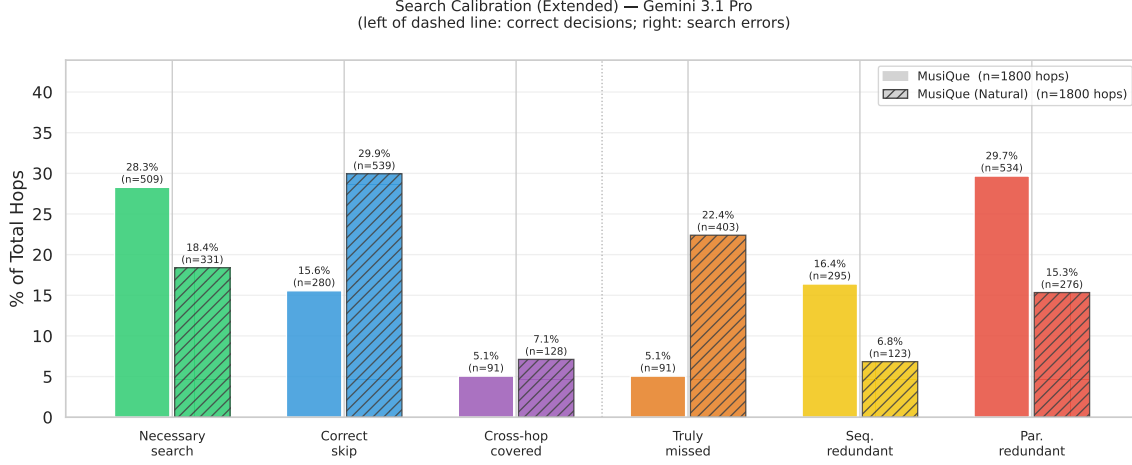


Figure 2: Extended search-calibration profile for Gemini 3.1 Pro. Solid bars: benchmark MuSiQue; hatched bars: natural paraphrase. Left of the dashed line are correct decisions (Necessary search, Correct skip, Cross-hop covered); right of the dashed line are the three error categories. The benchmark→natural shift moves mass from Necessary search into Truly missed (and from Par. redundant into Correct skip), exactly the calibration regression Table 4 reports. Plots for the other two models are in Figure 4.

- **Gemini** is a *high-volume, redundancy-prone* searcher. Its dominant error mode is *Parametrically redundant* (20.5%) plus *Sequentially redundant* (9.2%) — nearly 30% of its hops are wasted searches on content it already has, either in its parameters or in earlier turns of the trace. Coverage of uncertain hops, however, is excellent: *Truly missed* is only 5.1%.
- **Nemotron** is the inverse archetype — a *coverage-failing under-searcher*. Redundancy of both kinds is small (1.3% + 5.4%), but *Truly missed* dominates at 24.3%: nearly a quarter of all hops are uncertain and the model never even attempts a query.
- **Qwen** sits between the two. It searches more uncertain hops than the other two (49.0% *Necessary search*) but pays for it with 19.8% *Par. redundant*, mirroring Gemini’s redundancy pattern at slightly lower volume.

The three-way error decomposition is essential here. Any binary collapse to “correct search-decision” versus “error” would hide that Gemini and Nemotron fail in opposite ways. Cross-hop coverage — the case where an uncertain hop is recovered as a by-product of another hop’s query — is non-negligible (5.1–10.2%) for all three; we count it as a correct outcome, since the hop’s information need is in fact satisfied, but it is invisible in a strict “did the model search this hop” framing and would be misread as a miss without this annotation.

Cross-model statistical contrast. A Kruskal–Wallis test on the per-hop entropy distributions rejects the null of equal distributions across the three models ($H = 499.2$, $p < 10^{-108}$), confirming that the behavioral differences are not driven by trivially different uncertainty inputs. A Mann–Whitney U test on the entropy of *searched* hops shows Nemotron searches substantially more uncertain hops on average than Gemini ($p < 10^{-97}$), consistent with the very different positions of the two models on the redundancy/coverage axis.

5 Entropy Barely Predicts Search

If models routed search by their own uncertainty, the AUROC of entropy $\rightarrow$ search would approach 1.0. We observe instead AUROC in the 0.56–0.62 range across all three models (Figure 3). This is the central diagnostic of the paper: *the latent uncertainty signal exists — but it is not the dominant driver of when these models decide to search*.

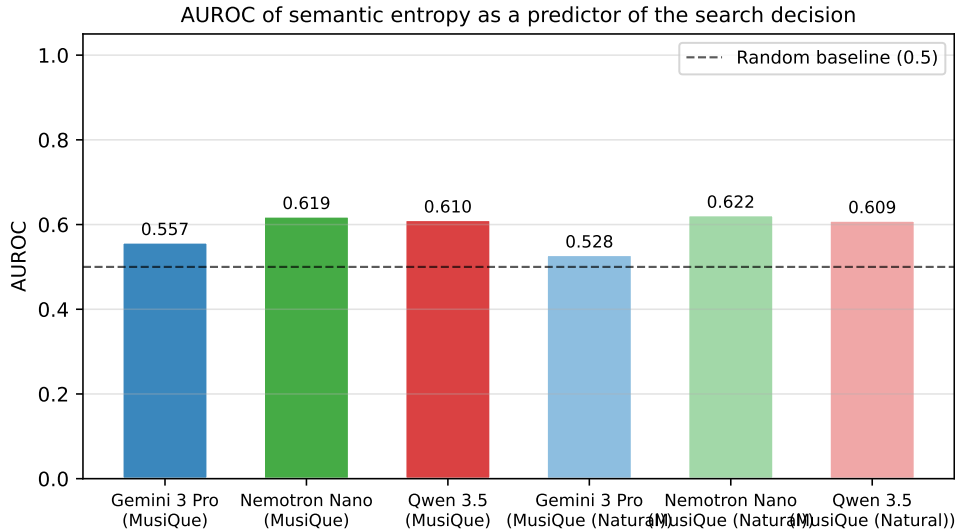


Figure 3: AUROC of semantic entropy as a predictor of the per-hop search decision, per (model, dataset). The dashed line marks the random baseline (AUROC = 0.5). All six combinations fall in the 0.53–0.62 range; for Gemini under natural paraphrasing, AUROC drops to 0.528 — effectively chance.

Two observations sharpen this finding. (1) Even the *best* of the three, Nemotron, clears AUROC 0.6 only modestly — and pays for that calibration with the highest *Truly missed* rate of the three (Table 2). (2) The qualitative shape of the per-entropy-level search rate is roughly flat for all three models rather than monotone — search is approximately *binary* (the model either searches a hop or doesn’t, with little graded response to the magnitude of internal uncertainty). The per-entropy-level rate plot supporting this binary-behavior claim is in Appendix B; the qualitative pattern it shows is consistent with the near-chance AUROC reported here.

6 Generalization Under Natural Paraphrases

The protocol so far evaluates models on benchmark questions, which in MuSiQue expose the hop chain syntactically (“What administrative territorial entity includes the place where X was educated?”). A natural user with the same information need would phrase the question differently (“I’m curious about X’s background — what region of the country did he study in?”). Two facts about this distinction motivate our second experiment: (i) the underlying information need is identical (same gold answer, same hop chain semantically); (ii) the surface form differs (no nested relative clauses, no exposed bridge entities, often a request for a short narrative rather than a single fact).

If the search-decision policy were governed by the model’s internal uncertainty about facts, behavior should be approximately invariant to this surface change. We test this directly.

Paraphrase generation. We use `gemma4:31b` to paraphrase each of the 600 MuSiQue questions into a natural, goal-oriented rewrite. The system prompt (Appendix E) enforces three verifiable constraints: (i) no intermediate bridge entity may be named, (ii) the “X of Y where Z did W” relative-clause pattern is forbidden, and (iii) phrasing must invite 2–4 explanatory sentences rather than a one-word answer. Semantic equivalence is guaranteed by construction: the same gold answers and the same hop decompositions from the

original questions are reused in Stage 3 attribution, so any divergence in the gold answer chain would surface immediately as attribution failure. Surface-form coverage is an established strength of large instruction-tuned models for this class of constrained rewrite (Jang et al., 2026; Cheng et al., 2026). We then re-run only Stages 2–3 (the search-augmented run and hop attribution); Stage 1 parametric uncertainty is *reused from the original questions* because the hop chain is semantically unchanged.

Result 1 — Search volume collapses. All three models cut search volume substantially (Table 3): Qwen by 4.2×, Gemini by 3.1×, Nemotron by 1.8×. The information need is identical; only the phrasing changed.

Table 3: Search volume per question, benchmark vs. natural paraphrase (paired across 600 items). Wilcoxon signed-rank test, two-sided.

Model	Benchmark (mean)	Natural (mean)	Ratio	Wilcoxon p
Gemini 3.1 Pro	15.20	4.93	3.1×	3.0×10^{-73}
Nemotron Nano 30B	5.74	3.13	1.8×	2.4×10^{-21}
Qwen 3.5 122B	5.60	1.33	4.2×	6.4×10^{-88}

Result 2 — Calibration collapses: the wrong cells grow. Reduced search would be unsurprising if the model’s calibration *survived*: a leaner search budget could still be allocated to uncertain hops. It does not. Table 4 shows the six-category shift from benchmark to natural paraphrase for each model. The pattern is uniform across models: *Necessary search* collapses by 11–27 percentage points, and the lost mass moves predominantly into *Truly missed* (uncertain hop the model fails to search at all). *Par. redundant* also drops — which would be a positive in isolation — but the trade-off is unfavourable: the model loses correct searches faster than it loses wasted ones.

Table 4: Six-category profile shift under natural paraphrasing (Δ = Natural – Benchmark, in percentage points of 1800 hops). All three models lose *Necessary search* and gain *Truly missed*.

Model	Correct		Errors			
	Δ Nec. srch.	Δ Corr. skip	Δ Truly miss.	Δ X-hop cov.	Δ Seq. red.	Δ Par. red.
Gemini 3.1 Pro	−19.4	+14.3	+17.3	+2.0	−6.1	−8.2
Nemotron Nano 30B	−10.5	+3.4	+9.2	+1.4	−0.7	−2.7
Qwen 3.5 122B	−27.4	+15.5	+27.7	−0.3	−2.4	−13.1

Result 3 — The calibration gap persists or widens. Under natural paraphrasing the AUROC of entropy $\rightarrow$ search drops marginally for Gemini (0.557 $\rightarrow$ 0.528) and is essentially unchanged for the two open models. The headline finding of §5 — that entropy is a weak predictor of search — *persists, and degrades for the highest-volume searcher*. Surface form, not internal uncertainty, governs the policy.

Synthesis. The two results together rule out the most charitable interpretation of the volume drop. If the model had become “smarter” under natural framing — searching less but routing better — we would see PR cells shrink while *Necessary search* held steady. We instead see both cells move in the wrong direction together: the categories that grow are *Truly missed* and *Correct skip*, exactly what one would expect from a model that has shifted its overall search prior downward irrespective of per-hop uncertainty. Two complementary probes in the appendix support a specific mechanism: under natural framing, models commit to per-hop answers *earlier in the trace, before any search is issued* (Appendix C), and the growth of *Truly missed* is dominated by genuine misses rather than by downstream cascade failures from a wrong earlier hop (Appendix D).

7 Discussion

Why does framing flip behavior so cleanly? The probes in Appendix C are consistent with the natural-paraphrased question being interpreted as a casual, narrative request rather than a structured

retrieval task. The model’s *search policy* then conditions on its read of the prompt’s register rather than on per-hop epistemic state. This is a behavioral analogue of the “context leads, parametric memory follows” effect documented by Tao et al. (2024) for *generation*: here it is the *search-decision* policy, not the answer, that the model conditions on surface context. Cheng et al. (2026) provide a mechanistic account: representation probing reveals that the model’s cognition layer (recognising that a tool call is needed) and its execution layer (emitting tool-call tokens) become nearly orthogonal under conversational framing — the model “knows” it should search but fails to act on that knowledge. Our commitment-locus probe (Appendix C) provides a behavioral correlate of this gap at hop granularity.

Implications for tool-use training. The standard RAG / tool-use training paradigm — Toolformer-style supervised insertion (Schick et al., 2023), Self-RAG critique tokens (Asai et al., 2024), or distillation from agentic trajectories (Wang et al., 2025b) — does not explicitly tie the decision to search to the model’s own uncertainty about the sub-step. Our finding suggests these methods may produce models whose search behavior generalizes poorly off the training distribution of question forms. A growing body of work addresses exactly this gap: QuCo-RAG (Min et al., 2025) injects search only when corpus statistics confirm a knowledge deficit; Unified Active Retrieval (Cheng et al., 2024b) trains a hidden-state classifier to gate retrieval on per-step uncertainty; R3-RAG (Li et al., 2025) combines outcome reward with a step-level process reward for intermediate document relevance; and S-GRPO (Suri et al., 2025) rewards search only when the model’s internal belief state warrants it, boosting a 3B model from 36.5% to 65.2% on a when-to-call benchmark. Our evaluation protocol provides a *diagnostic* that complements these training approaches: by measuring the calibration gap between parametric uncertainty and search decisions on a held-out set, it can quantify whether a given training method actually closes the miscalibration we document.

Limitations. (i) Per-hop attribution uses an LLM judge; we mitigate this with high-confidence-only ablations and substring-fallback checks, and the qualitative pattern is robust. (ii) 600 questions is large enough for cross-condition contrasts but constrains finer slicing (e.g. by hop-count $\times$ topic). (iii) Semantic entropy at $N = 5$ admits only 7 discrete values; finer probes (Kunitomo-Jacquin et al., 2026) may sharpen the calibration metrics. (iv) We do not vary the search backend; results may differ with stronger retrieval. (v) Only one paraphrase per question is evaluated; the generalization gap may be even larger across the full natural-language manifold.

8 Conclusion

Info-seeking queries put parametric memory and live web search in continuous tension — one is fast, free, and increasingly capable; the other is current, grounded, and increasingly easy to invoke. Existing benchmarks tell us whether RAG helps on aggregate; they do not tell us whether a model knows *when* to defer to its parameters and when to reach for the web. We introduce a protocol that asks this question directly, grounded in per-hop semantic entropy and per-hop search attribution, and use it to profile three production-style LLMs through a six-category extended calibration view that distinguishes three correct decisions (*Necessary search*, *Correct skip*, *Cross-hop covered*) from three error modes (*Truly missed*, *Sequentially redundant*, *Parametrically redundant*). The profiles reveal qualitatively different parametric $\leftrightarrow$ search archetypes across models; the AUROC of entropy $\rightarrow$ search confirms that on these info-seeking queries search behaviour is at best weakly aligned with the model’s own internal knowledge boundary; and a single cosmetic paraphrase — same information need, same gold answer — shifts every model toward the worst error cell, *Truly missed*. We argue this constitutes a hidden generalization gap in the parametric $\leftrightarrow$ search interplay of contemporary info-seeking LLMs and that the protocol introduced here is a starting point for measuring and ultimately closing it as both axes of model capability — parametric knowledge and agentic tool use — continue to advance.

References

A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *ICLR*, 2024.

- S. Cheng, L. Pan, X. Yin, X. Wang, and W. Y. Wang. Understanding the interplay between parametric and contextual knowledge for large language models. *arXiv:2410.08414*, 2024.
- M. Farahani and R. Johansson. Deciphering the interplay of parametric and non-parametric memory in retrieval-augmented language models. *EMNLP*, 2024.
- S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.
- Y. Huang and J. X. Huang. A survey on retrieval-augmented text generation for large language models. *ACM Computing Surveys*, 2024.
- Y. Kim et al. Reliability across parametric and external knowledge: Understanding knowledge handling in LLMs. *arXiv:2502.13648*, 2025.
- L. Kunitomo-Jacquin et al. Evidential semantic entropy for LLM uncertainty quantification. *EACL*, 2026.
- J. Lee and J. Hockenmaier. Evaluating step-by-step reasoning traces: A survey. *arXiv:2502.12289*, 2025.
- Z. Li et al. Understanding and patching compositional reasoning in LLMs. *Findings of ACL*, 2024.
- S. Min et al. FActScore: Fine-grained atomic evaluation of factual precision in long-form text generation. *EMNLP*, 2023.
- K. Ning et al. WTU-Eval: A whether-or-not tool usage evaluation benchmark for large language models. *arXiv:2407.12823*, 2024.
- T. Schick et al. Toolformer: Language models can teach themselves to use tools. *NeurIPS*, 2023.
- Y. Tao et al. When context leads but parametric memory follows in large language models. *EMNLP*, 2024.
- H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal. MuSiQue: Multihop questions via single-hop question composition. *TACL*, 10:539–554, 2022.
- H. Wadhwa et al. From RAGs to rich parameters: Probing how language models utilize external knowledge over parametric information for factual queries. *WSDM*, 2026.
- Z. Wang, A. Mao, H. Wu, T. Bui, J. Wang, Y. Tang, Y. Li, Y. Zhuang, J. Kang, S. C. H. Hoi, and W. Wang. RARE: Retrieval-augmented reasoning modeling. *arXiv:2503.23513*, 2025.
- K. Wu, E. Wu, and J. Zou. ClashEval: Quantifying the tug-of-war between an LLM’s internal prior and external evidence. *NeurIPS*, 2024.
- J. Wu et al. MRKE: The multi-hop reasoning evaluation of LLMs by knowledge edition. *arXiv:2402.11924*, 2024.
- S. Yang et al. Do large language models latently perform multi-hop reasoning? *ACL*, 2024.
- H. Zhang et al. Evaluating the external and parametric knowledge fusion of large language models. *arXiv:2405.19010*, 2024.
- Z. Jiang, F. Xu, L. Gao, Z. Sun, Q. Liu, J. Dwivedi-Yu, Y. Yang, J. Callan, and G. Neubig. Active retrieval augmented generation. *EMNLP*, 2023.
- X. Guan, J. Zeng, F. Meng, C. Xin, Y. Lu, H. Lin, X. Han, L. Sun, and J. Zhou. DeepRAG: Thinking to retrieve step by step for large language models. *arXiv:2502.01142*, 2025.
- P. Wu, M. Zhang, Z. Sheng, S. Wang, X. Du, and Z. Chen. HiPRAG: Hierarchical process rewards for efficient agentic retrieval augmented generation. *arXiv:2510.07794*, 2025.
- D. Guo, J. Wu, and S. M. Yiu. When to retrieve during reasoning: Adaptive retrieval for large reasoning models. *arXiv:2604.26649*, 2026.
- C. Qian, E. C. Acikgoz, H. Wang, X. Chen, A. Sil, D. Hakkani-Tür, G. Tur, and H. Ji. SMART: Self-aware agent for tool overuse mitigation. *Findings of ACL*, 2025.
- P. Wu, M. Zhang, X. Zhang, X. Du, and Z. Chen. Search wisely: Mitigating sub-optimal agentic searches by reducing uncertainty. *EMNLP*, 2025.
- Y. Jang et al. How you ask matters! Adaptive RAG robustness to query variations. *arXiv:2604.10745*, 2026.
- Y. Cheng, C. Fan, M. JafariRaviz, K. Rezaei, and S. Feizi. Model-adaptive tool necessity reveals the knowing-doing gap in LLM tool use. *arXiv:2605.14038*, 2026.
- M. Rabinovich, A. Anaby-Tavor, et al. On the robustness of agentic function calling. *TrustNLP workshop, NAACL*, 2025.
- H. Wang, C. Qian, Y. Liu, J. Liu, Z. Du, M. He, B. Hu, K.-F. Wong, and H. Ji. Position: Agent should invoke external tools only when epistemically necessary. *ICML Position Track*, 2025.

- Y. Wang, L. Wei, and H. Ling. Retrieval as a decision: Training-free adaptive gating for efficient RAG. *TMLR*, 2026.
- H. Zubkova, J.-H. Park, and S.-W. Lee. SUGAR: Leveraging contextual confidence for smarter retrieval. *ICASSP*, 2025.
- Z. Yao et al. SeaKR: Self-aware knowledge retrieval for adaptive retrieval augmented generation. *ACL*, 2025.
- SEARAG: Semantic entropy-guided adaptive retrieval for multi-hop question answering. *OpenReview submission* <https://openreview.net/forum?id=suKWySDjil>, 2025.
- J. Kossen et al. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv:2406.15927*, 2024.
- He, Hu et al. MINTQA: A multi-hop question answering benchmark for evaluating LLMs on new and tail knowledge. *arXiv:2412.17032*, 2024.
- H. Han et al. Judge’s verdict: A comprehensive analysis of LLM judge capability through human agreement. *arXiv:2510.09738*, 2025.
- H. Wang, A. Prasad, E. Stengel-Eskin, and M. Bansal. AdaCAD: Adaptively decoding to balance conflicts between contextual and parametric knowledge. *NAACL*, 2025.
- Khandelwal et al. CoCoA: Confidence- and context-aware adaptive decoding for resolving knowledge conflicts in LLMs. *EMNLP*, 2025.
- Huo et al. Micro-Act: Mitigate knowledge conflict in question answering via actionable self-reasoning. *ACL*, 2025.
- D. Min, N. Zhang, Y. Yao, X. Wang, J. Du, S. Deng, H. Chen, and J. Pujara. QuCo-RAG: Quantifying uncertainty from the pre-training corpus for dynamic retrieval-augmented generation. *arXiv:2512.19134*, 2025.
- Q. Cheng, X. Li, S. Li, Q. Zhu, Z. Yin, Y. Shao, L. Li, T. Sun, H. Yan, and X. Qiu. Unified active retrieval for retrieval augmented generation. *Findings of EMNLP*, 2024.
- Y. Li, Q. Luo, X. Li, B. Li, Q. Cheng, Y. Zheng, Z. Yin, P. Wang, B. Yu, and X. Qiu. R3-RAG: Learning step-by-step reasoning and retrieval for LLMs via reinforcement learning. *Findings of EMNLP*, 2025.
- M. Suri, P. Mathur, N. Lipka, F. Dernoncourt, R. A. Rossi, and D. Manocha. Structured uncertainty guided clarification for LLM agents. *arXiv:2511.08798*, 2025.

A Extended Search-Calibration Plots for Nemotron and Qwen

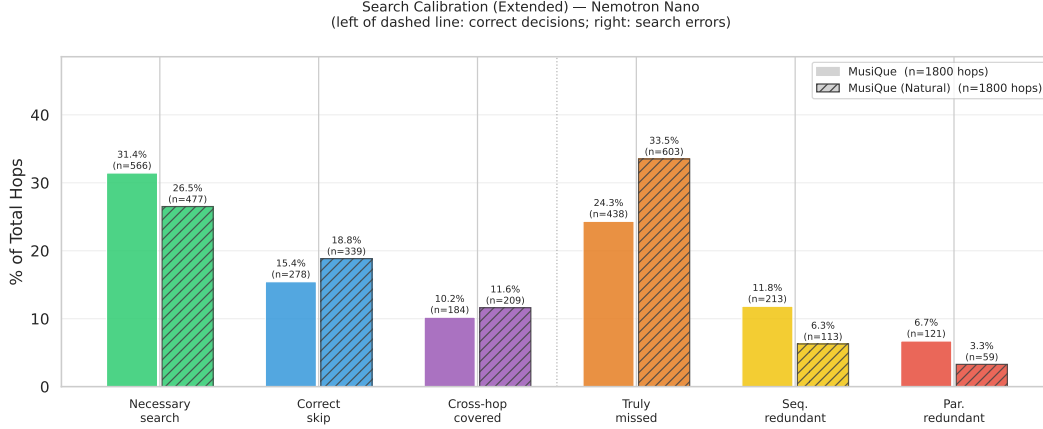
Per-model extended-calibration plots for the two models not shown in §4. The bars use the same colour code as Figure 2: green / blue / purple on the left of the dashed line are correct decisions; orange / yellow / red on the right are the three distinct error categories.

B Binary Search Behavior across Entropy Levels

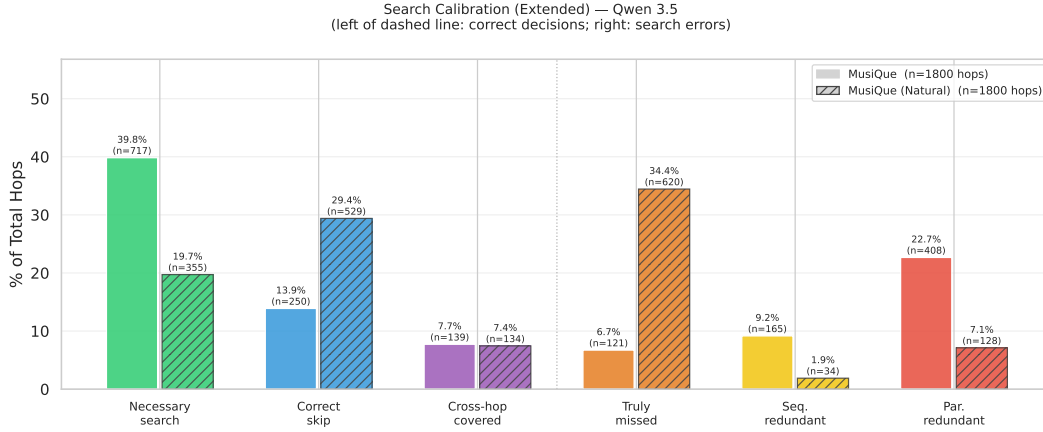
A direct calibration view plots the search rate $P(\text{searched} \mid H = h)$ as a function of the discrete entropy level h produced by $N = 5$ closed-book sampling. Under well-calibrated behavior, this curve would rise monotonically from $H=0$ (the model should rarely search; it knows the answer) to $H=2.32$ (the model should almost always search; it has near-uniform disagreement across samples). What we observe instead (Figure 5) is a roughly flat curve for all three models: the search rate at $H=0$ is only a little below the rate at the highest entropy levels, and the intermediate levels are noisy. A χ^2 test of uniformity across entropy levels does reject the null for every model ($p < 10^{-4}$), so a small signal is statistically present — but the practical shape is binary: the model either searches at a roughly-constant rate or it doesn’t, with little graded sensitivity to how uncertain it is. This is the per-entropy view of the near-chance AUROC reported in §5.

C Commitment-Locus Probe

Motivation. If natural framing causes models to skip searches they would otherwise have issued, *when* does the skip happen? One hypothesis is that the casual register of the paraphrased question invites a quick



(a) Nemotron Nano 30B — coverage-failing under-searcher: *Truly missed* dominates (24.3% on the benchmark, 33.5% on natural), while both redundancy categories stay small.



(b) Qwen 3.5 122B — the most aggressive volume drop under natural paraphrasing (search rate 4.2× lower), with *Truly missed* expanding from 6.7% to 34.4%.

Figure 4: Extended search-calibration profiles for the remaining two models.

narrative answer, leading the model to commit to a per-hop answer early — before any search query is issued for that hop. The *commitment-locus probe* tests this directly.

Probe. For every (example, hop, variant) appearing in both the benchmark trace and the natural trace of the same model, an LLM judge (`gpt-oss:120b`) identifies the first turn at which the agent commits to a specific per-hop answer (asserts “X is Y” or treats Y as established and uses it downstream). Combined with the first turn at which a search query is attributed to that hop, this yields `committed_before_search` per (hop, variant). The decisive cell is `commit_no_search`: the agent committed to an answer for that hop and no search was ever attributed to it.

Result. Table 5 reports the headline rates; Figure 6 visualizes them. Across all three models, the rate of pre-search commitment rises substantially under natural framing. For Gemini, commitment-without-search jumps from 22.6% of hops on the benchmark to 51.9% on the natural rewrite. For Qwen the rise is 18.0% → 38.5%. Nemotron, which already searched the least on the benchmark, shows a smaller absolute increase.

Interpretation. The shift in commitment locus is a mechanistic explanation for the *Truly missed* growth reported in §6: under natural framing the model is not simply “forgetting” to search — it is *deciding*

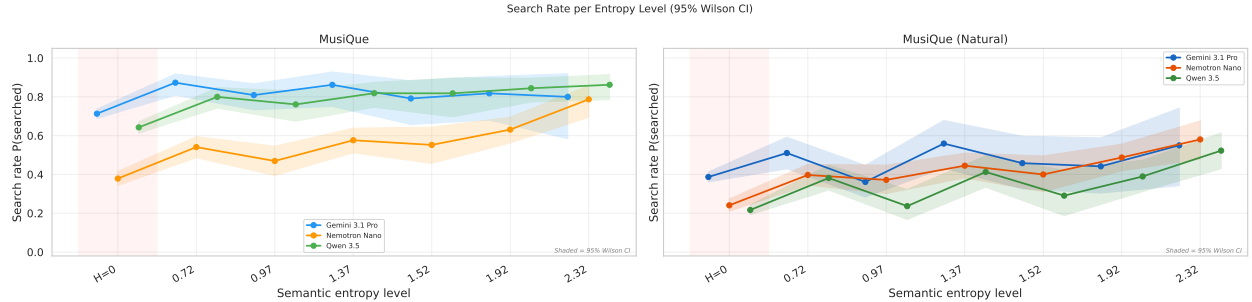


Figure 5: Search rate $P(\text{searched} | H = h)$ per entropy level, per model, on MuSiQue (left panel) and MuSiQue (Natural) (right panel). A well-calibrated policy would rise monotonically from $H=0$ to $H=2.32$. Observed curves are roughly flat on both datasets: behavior is approximately binary rather than graded with internal uncertainty. Shaded band = 95% Wilson CI.

Table 5: Commitment-locus rates per (model, variant). **commit_before_search**: agent committed before any search query attributed to the hop. **commit_no_search**: agent committed and no search ever attributed.

Model	Variant	Committed at all	Before search	No search at all
Gemini 3.1 Pro	Benchmark	78.5%	34.7%	22.6%
	Natural	56.1%	52.2%	51.9%
Nemotron Nano	Benchmark	52.2%	22.1%	21.5%
	Natural	54.0%	28.1%	27.8%
Qwen 3.5 122B	Benchmark	75.9%	19.6%	18.0%
	Natural	61.7%	38.9%	38.5%

the answer earlier, often before any search would be triggered. This is consistent with the “context leads, parametric memory follows” phenomenon of [Tao et al. \(2024\)](#), re-cast at the level of the *search-decision* policy rather than the answer-generation policy.

D Missed-Hop Confounder Analysis

Motivation. *Truly missed* is the cell that grows the most under natural paraphrasing (Table 4). A potential confound is that some of these misses are not the model’s failure to act on its own uncertainty but downstream casualties of an earlier mistake: if hop k resolves to a wrong entity, the model has nothing to anchor a query for hop $k+1$. We separate the two.

Decomposition. For each hop in the *Truly missed* cell, we classify:

- **genuine_missed**: $\text{hop_index} = 0$, or every prior hop’s parametric runs were all correct. The model could have searched and chose not to.
- **cascade_possible**: $\text{hop_index} > 0$ and at least one prior hop had parametric accuracy < 1.0 . The model may have been “starved” of the anchor entity.

Result. Table 6 and Figure 8 show that the growth in *Truly missed* under natural paraphrasing is dominated by the *genuine_missed* component for all three models. Genuine misses roughly triple for Gemini (84 $\rightarrow$ 277) and triple-plus for Qwen (105 $\rightarrow$ 369); Nemotron rises by 42% (240 $\rightarrow$ 342). This rules out the trivial explanation that the natural-paraphrase generalization gap is a downstream casualty of upstream error.

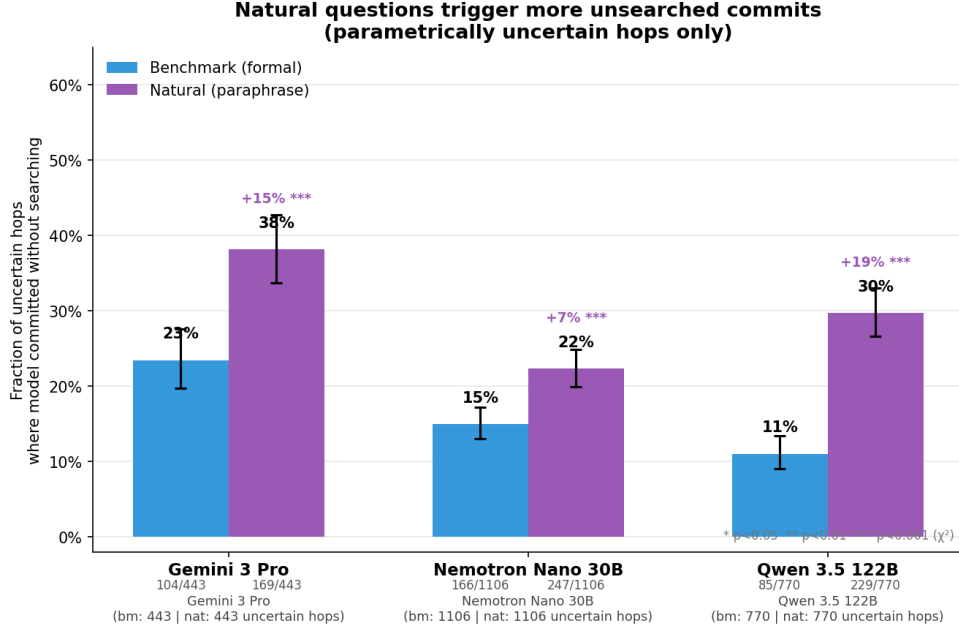


Figure 6: Pre-search commitment rate per (model, variant). Natural framing materially shifts the locus of the per-hop answer to *before* any search query is issued.

Table 6: *Truly missed* decomposition: genuine misses vs. cascade-possible misses, per (model, variant). Counts are of hops in the missed cell.

Model	Variant	Cascade possible	Genuine missed
Gemini 3.1 Pro	Benchmark	98	84
	Natural	254	277
Nemotron Nano	Benchmark	382	240
	Natural	470	342
Qwen 3.5 122B	Benchmark	155	105
	Natural	385	369

E Prompts

All LLM-judge calls in the protocol use the prompts reproduced below verbatim. Curly-brace placeholders are filled at runtime with the per-example fields named in the text.

E.1 Staleness Judge (Stage 0)

Model: gpt-oss:20b via Ollama.

System. You are an expert at identifying whether a factual question requires time-sensitive knowledge.

A question is STALE if its correct answer could plausibly change over time. This includes:

- Current officeholders (presidents, ministers, CEOs, etc.)
- Current statistics (population, GDP, rankings, prices)
- Current records (tallest building, richest person, etc.)
- Ongoing relationships (current spouse, current employer, current team)
- Any fact described as "current", "present", "now", "today", or "latest"

A question is NOT STALE if its correct answer is a fixed historical fact:

- Historical events and their dates
- Births, deaths, and biographical facts about historical figures
- Geographic facts (where a place is located, what country something is in)

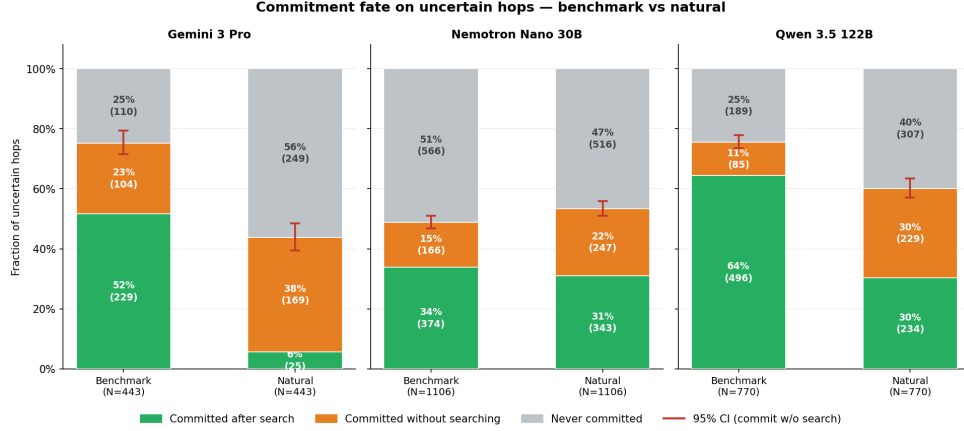


Figure 7: Commitment-fate breakdown per (model, variant). The growth of the orange “committed without searching” band under natural framing is the per-model picture of the generalization gap localized to a single behavioral mechanism.

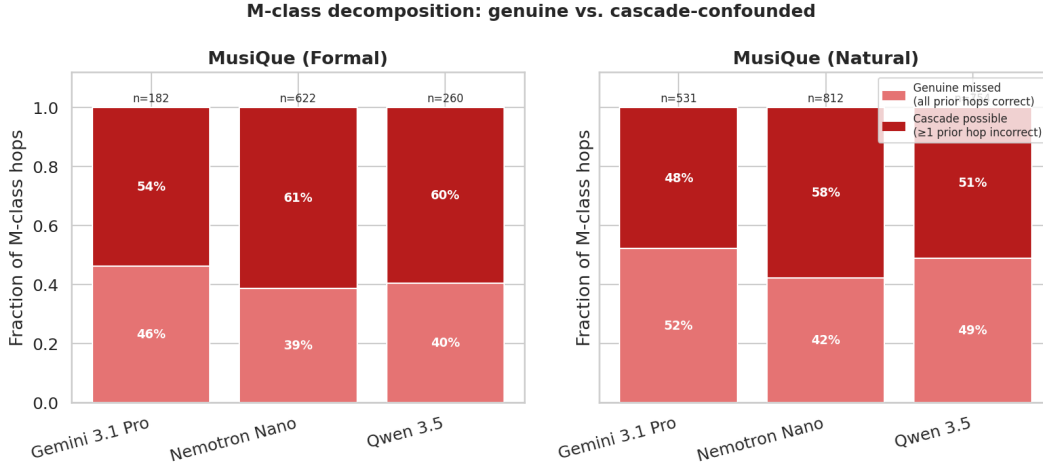


Figure 8: *Truly missed* cell decomposition. The dark (genuine_missed) component grows under natural framing for every model, ruling out cascade-driven explanations for the missed-cell expansion.

- Authorship, invention, or creation of things
- Past achievements and awards
- Scientific or mathematical constants

For multi-hop questions, mark as STALE if ANY hop requires time-sensitive knowledge.

User. Classify the following question as STALE or NOT_STALE.

Question: {question}

Think through whether any part of answering this question requires knowing time-sensitive information (facts that could change over time, like current officeholders, current statistics, or ongoing relationships).

Respond in exactly this format:

Classification: STALE or NOT_STALE

Reason: <one sentence explaining why>

E.2 Semantic Clusterer (Stage 1)

Model: gpt-oss:120b via Ollama. The structured output is a list of integer cluster IDs of length $N = 5$.

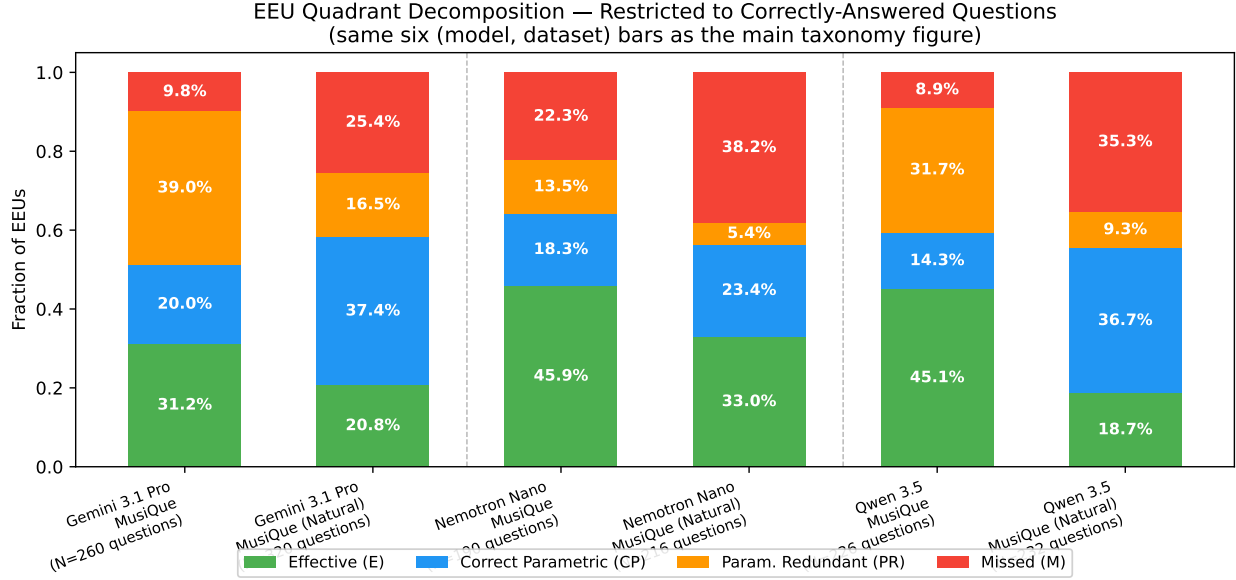


Figure 9: **Robustness check.** Reproduction of the headline taxonomy figure (Figure 1) restricted to questions on which the model produced the correct final aggregate answer. Even within this favourable sub-population, the natural-paraphrase bars show the same expansion of the red (*Missed*) segment and contraction of the green (*Effective*) segment. The calibration-and-paraphrase phenomena documented in the main text are therefore not artefacts of failure-mode questions: they appear inside the subset of questions the model actually got right.

```

Question: {question}

The following {n} answers were given independently to this question:
{answers}

Group them by semantic equivalence. Two answers belong to the same cluster if they express the
same fact (minor wording differences are fine; "United States" and "USA" are the same). Use
consecutive integer cluster IDs starting from 0.

```

E.3 Paraphrase Rewriter (Paraphrase generation)

Model: gemma4:31b via Ollama.

System. You are a prompt rewriter. You will be given a multi-hop benchmark question and the chain of sub-questions that reveal its structure. Your task is to rewrite the question so it sounds like a genuine, curious user asking -- not like a database traversal.

Rules:

1. Do NOT mention any intermediate entity that appears only as a bridge in the hop chain. The user asking the question would not know that entity exists.
2. Do NOT use the "X of Y where Z did W" pattern. Avoid nested relative clauses that signal the hop structure.
3. Frame the question so the answer naturally warrants 2-4 explanatory sentences, not a single word or number. Prefer "why", "how", "tell me about", "explain", "what can you tell me about" phrasing where it fits naturally. If the answer is inherently factual (a location, a date), still frame it as "I'm curious about..." or "Could you tell me a bit about..." to invite context.
4. The rewritten question must preserve the same ultimate information need and have the same gold answer.
5. Return only the rewritten question text -- no explanation, no preamble, no quotes.

User. Original benchmark question: {question}

Hop chain (do NOT expose these steps in the rewrite):
{hops}

Gold answer: {answer}

Rewrite this as a natural user question following the rules:

E.4 Query→Hop Attribution Judge (Stage 3)

Model: gpt-oss:120b via Ollama. The structured output is the integer hop index plus a confidence label.

```
You are attributing a search query issued by an AI agent to one of the sub-questions (hops) that
make up a multi-hop question.
Aggregate question: {aggregate_question}
Sub-questions (hops):
{sub_questions_text}
Agent's thinking before the search query:
{thinking}
Search query: {query}
Which sub-question (hop) is this search query primarily trying to answer?
Return the 0-based hop_index (0, 1, 2, ...) or -1 if the query is about the aggregate question or
doesn't clearly target a specific sub-question.
```

E.5 Commitment-Locus Judge (Appendix C)

Model: gpt-oss:120b via Ollama.

```
You are analyzing a transcript of an AI agent solving a multi-hop question by interleaving
reasoning, search-tool calls, and search results. Each turn in the transcript is prefixed with
[TURN i: ROLE ...].
Aggregate question: {aggregate_question}
Target sub-question (one hop of the chain): {hop_question}
Gold answer for this sub-question: {gold_answer}
Your task: identify the FIRST turn at which the agent commits to a specific answer for the TARGET
SUB-QUESTION above (not the aggregate question and not any other hop).
"Commits" means:
- Asserts a definite answer ("X is Y", "the answer is Z"), OR
- Treats an answer as established and reuses it in subsequent reasoning (e.g. forms the next hop's
search query as if the prior answer is settled).
If a turn merely lists candidate answers, expresses doubt, or fetches information without
concluding, it is NOT a commitment. The very first turn that satisfies the above is the
commitment turn.
Notes:
- Turn indices are 0-based and match the [TURN i: ...] prefix.
- Return -1 for first_committed_turn_index if no commitment occurs anywhere in the transcript.
- matches_gold compares the COMMITTED answer to the GOLD answer; commitment exists even when wrong.
==== TRANSCRIPT ====
{transcript}
==== END TRANSCRIPT ====
```